

CSE 4408

System Analysis and Design

Lab 7: Agile Modeling, Prototyping, and Scrum

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MedRadar

by team **CookiesAndCaches**

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Stakeholder Needs & Derived Features

- **Patient/Customers:** Need a quick and reliable way to locate nearby pharmacies and check real-time medicine availability.
- **Pharmacy Owners:** Need an easy-to-use platform to update stock and attract more customers through increased visibility.

Feature List derived from the gathered information in Lab 5 and 6:

- User login and registration
- Search for medicine by categories and filters
- Pharmacy search by location and medicine availability
- Pharmacy stock dashboard with update function
- Pharmacy portal with restock alerts and sales insights
- Order medicine if delivery is possible
- Customer Feedback form

Prioritized User Stories

1. User Authentication & Login/Register System:

*As a customer,
I want to verify that pharmacies and other users are registered and legitimate, so I can trust the information and avoid falling victim to fraud.*

Priority: Must

Justification: Verifying users and pharmacies ensures reliable information and keeps the platform secure.

2. Medicine Availability Info:

*As a customer,
I want to check the availability of specific medicines, so that I avoid visiting pharmacies that are out of stock.*

Priority: Must

Justification: Critical for users to make informed decisions before traveling or placing an order.

Prioritized User Stories(Cond.)

3.Pharmacy Location Search:

*As a customer,
I want to search for pharmacies near me
that have my required medicine, so that I
can find one close to me quickly.*

Priority: Must

Justification: Core functionality;
addresses the primary user need of
locating nearby pharmacies

4. Pharmacy Stock Management

*As a pharmacy owner,
I want to update my stock information
regularly, so that customers see accurate
availability.*

Priority: Must

Justification: Keeps stock data reliable
and up-to-date.

Prioritized User Stories(Cond.)

5. Interactive Map View:

As a customer, I want to see pharmacies displayed on a map, so that I can easily navigate to them.

Priority: Could

Justification: Improves user experience, but not mandatory initially.

6. Order Placement with Pickup or Delivery Option

As a patient, I want to place an order for medicines and choose pickup or delivery, so that I can get my medicines conveniently.

Priority: Should

Justification: Adds convenience and expands service reach; important, but can follow basic search and stock viewing.

Prototype

Type of Prototype:

Non-operational UI-UX prototype

Selected Prototype features :

1. User authentication and Login/Register system
2. Customer dashboard + Search Medicine

Prototype

Agile Planning – Scrum Roles

Role	Assigned Member	Justification
Product Owner	Mahdi	• Identifies user needs and extracts project features • Led user information gathering in Lab 5 and 6 to meet expectations
Scrum Master	Maisha	• Skilled at organizing previous labs and managing group workflows, • Keeps the team on track with processes and deadlines.
Dev Team Member	Obidit	• Has experience in UI design • Maintained technical documentation

Product Backlog Table

User	User Story Summary	Priority	Size Estimate
Customer	User Authentication for reliability	Must	M
Customer	Check the availability of specific medicines	Must	S
Customer	Search for pharmacies near me that have the required medicine	Must	M
Pharmacy Owner	To update stock information regularly	Must	L
Customer	Place orders with a pickup or delivery option	Should	L
Customer	Display pharmacies on an interactive map	Could	L

Sprint Planning

Selected User Stories for the first Sprint (2 weeks)

1. User Authentication– Login and Register System
2. Check for Medicine Availability Info
3. Nearby Pharmacy Location Search for having required Medicine

Overall Justification:

1. Core Functionality, all Must-haves
2. Highly Prioritized by Users; unlocks future modules
3. Small to Medium Complexity
4. Provides immediate value to users
5. Features 2 and 3 are connected, and it's better to implement both in one go

Agile vs SDLC reflection

Aspect	Agile (Our Approach)	SDLC (Alternative)
Requirements	Grows with feedback each sprint	Fully defined before starting
User Involvement	Active during analysis (surveys, interviews)	Mostly during initial phase
Flexibility	Can adapt to new needs (e.g., add order option)	Changes are slow and formal
Progress Style	Iterative, sprint by sprint	Linear, phase-by-phase
Delivery Speed	MVP and early features delivered quickly	Full system only at the end
Project Challenges	Frequent changes suit Agile's flexibility	Better if strict documentation is required early

Agile vs SDLC Reflection

Conclusion:

1. Helps refine and add features, aligning with user needs and expectations
2. Support quick changes to the initial plan
3. Fits our team's goal to validate ideas before full formal development

That's why Agile fits our project better than SDLC.

Thank you!

Feel free to ask any questions!